

BEACONHOUSE NATIONAL UNIVERSITY



LAB 10

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Total Marks: /20

Due Date: 28 Apr, 2025

Lab Manual

For

COAL(SEC_B)

Course Instructor	Ms. Nimra Abbas
Lab Instructor	Ayesha Seher
Semester	Spring 2025

Task 1: Implement bubble sort to arrange the elements of the given array temp in ascending order, using conditional jumps and loops based on the comparison of array values.

temp dw 60, 55, 45, 50, 40, 35, 25, 30, 10, 0 ; apply bubble sort in ascending order.

Read the sorted array from the emulator screen, and paste the code along with the output and Variables .

window showing the final sorted array.

Solution:

```
org 100h
.model small
.stack 100h
.data
    jmp start
    temp dw 60,55,45,50,40,35,25,30,10,0

.code
start:
main proc
    mov ax,@data
    mov ds,ax

    mov dx,9

loop1:
    mov cx,dx
    mov bx,offset temp
loop2:
    mov ax,[bx]
    cmp ax,[bx+2]
    jbe noswap

    xchg ax,[bx+2]
    mov [bx],ax
```

```

noswap:
add bx,2
loop loop2

```

```

dec dx
cmp dx,0
jne loop1

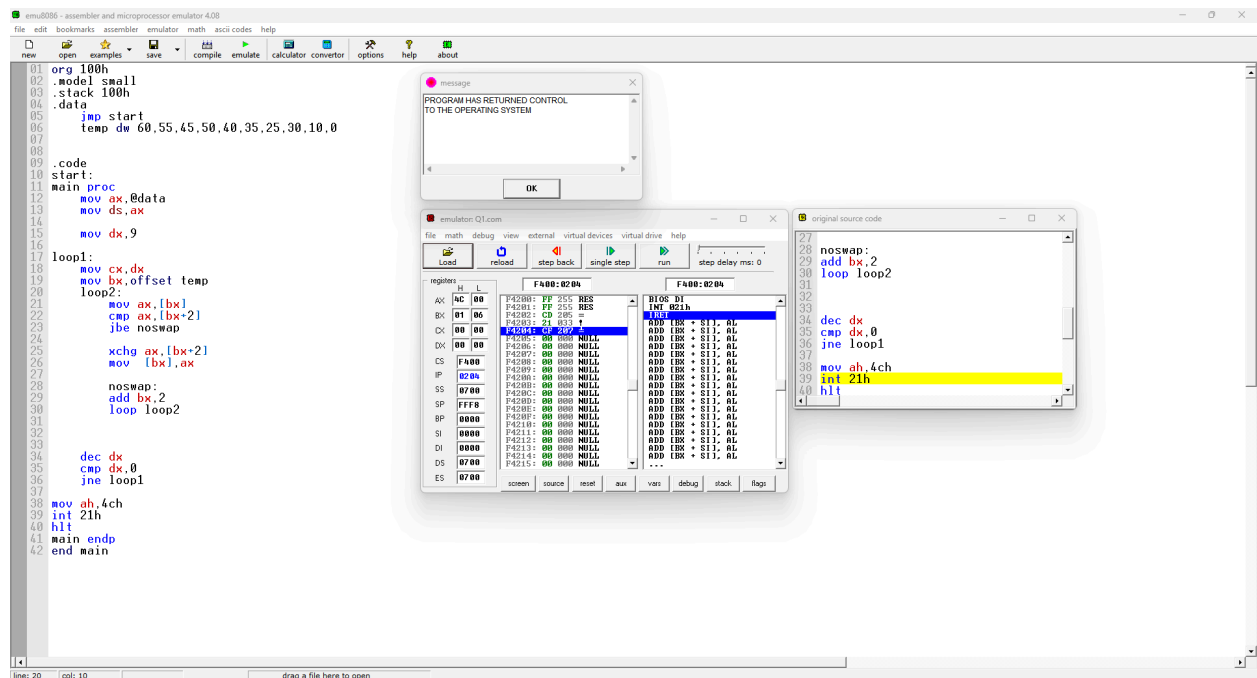
```

```

mov ah,4ch
int 21h
hlt
main endp
end main

```

Screenshot:



Task 1: Adds ten numbers without using a separate counter. Use conditional and unconditional jumps to control the flow of the program.

Solution:

```
org 100h
.model small
.stack 100h
.data
    num:  dw 10,20,30,40,50,10,20,30,40,50
    total: dw 0
.code
main proc
    mov ax, @data
    mov ds, ax

    mov bx, 0
    mov ax, 0

label: add ax, [num + bx]
        add bx, 2
        cmp bx, 20
        jne label

    mov [total], ax

    mov ah, 4ch
    int 21h

    main endp
end main
ret
```

Screenshot:

